

Closing the Count Gap in Quicksort

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Abstract

Randomized Quicksort has a discrete comparison-count law at every finite size, but its limiting law is continuous. We formalize a way to measure the finite-size gap in bits, give a construction that closes it exactly, and explain why that construction does not recover the tree’s dynamics. A finite-size split model and an error bound show how to pursue a useful, compact approximation instead.

From a good curve to an evolving model

Let C_n count comparisons when uniform-pivot Quicksort sorts n distinct keys. Its exact law obeys

$$C_n \stackrel{d}{=} n - 1 + C_{I_n} + C'_{n-1-I_n}, \quad I_n \sim \text{Uniform}\{0, \dots, n-1\}. \quad (1)$$

Our companion paper proves that an ordinary lognormal earns a higher expected log-density than an ordinary normal for every $n \geq 4$. Yet, after centering and scaling, C_n approaches a non-normal Quicksort law Q ; the fitted lognormal approaches a normal shape. A curve can therefore win at each finite size under one score and still miss the eventual shape.

The next question is how that shape develops. Write $\sigma^2 = 7 - 2\pi^2/3$ and $g(u) = 1 + 2u \log u + 2(1-u) \log(1-u)$. Starting at $\nu_0 = \delta_0$, apply the limiting split transform

$$\nu_{k+1} = \mathcal{T}(\nu_k), \quad \mathcal{T}(\nu) = \mathcal{L}(UX_1 + (1-U)X_2 + g(U)/\sigma), \quad X_1, X_2 \stackrel{\text{iid}}{\sim} \nu, \quad U \sim \text{Unif}(0, 1). \quad (2)$$

These centered laws approach Q , and $\text{Var}(\nu_k) = r_k := 1 - (2/3)^k$. A fractional generation is $\nu_{k,\alpha} = (1-\alpha)\nu_k + \alpha\nu_{k+1}$ with variance $r_{k,\alpha} = (1-\alpha)r_k + \alpha r_{k+1}$. For input size n , our smooth count model is

$$X_{n,k,\alpha} = \mu_n + s_n V_{k,\alpha} / \sqrt{r_{k,\alpha}}, \quad V_{k,\alpha} \sim \nu_{k,\alpha}, \quad \mu_n = \mathbb{E}C_n, \quad s_n^2 = \text{Var}(C_n). \quad (3)$$

It has the exact mean and variance of C_n , while its shape changes with recursive generation. The fitted interpolation describes central probabilities well but understates early-size skewness and the long right tail.

A finite bit measure for the count distribution

Write $p_n(c) = \Pr(C_n = c)$ for the exact probability of integer count c . Let $F_{n,\theta}$ be the CDF of a continuous approximation expressed in *comparison-count units*. Ask the same question of both models by rounding the continuous output to the nearest integer:

$$q_{n,\theta}(c) = F_{n,\theta}(c + \tfrac{1}{2}) - F_{n,\theta}(c - \tfrac{1}{2}). \quad (4)$$

Now p_n and $q_{n,\theta}$ live on the same count scale. With $m = (p_n + q_{n,\theta})/2$, their Jensen–Shannon divergence in bits is

$$\text{JS}_2(p_n, q_{n,\theta}) = \tfrac{1}{2} D_{\text{KL},2}(p_n \| m) + \tfrac{1}{2} D_{\text{KL},2}(q_{n,\theta} \| m). \quad (5)$$

It is zero exactly when the rounded model gets every count probability right. Operationally, it is how much one observed count reveals about whether it came from Quicksort or the model, assuming either source was equally likely.

Input size n	Fractional interpolation	Finite-size split correction
16	0.00509	0.00186
32	0.00235	0.00135
64	0.00085	0.00070

Two independent Monte Carlo runs; Jensen–Shannon bits per rounded count.

Reading the table. At $n = 16$, restoring finite-size structure cuts this mismatch by about 63%. The $n = 16$ correction uses exact child laws through size 8; the $n = 32, 64$ corrections use moment-matched Q child laws. The numbers are exploratory Monte Carlo estimates, especially close at $n = 64$.

An exact closure that teaches us what the score means

Let $U_n \sim \text{Unif}(-\frac{1}{2}, \frac{1}{2})$ be independent of the exact count C_n , and put $Y_n = C_n + U_n$. This Y_n has a continuous density, yet

$$\text{round}(Y_n) = C_n \quad \text{almost surely}, \quad q_n^Y(c) = p_n(c), \quad \text{JS}_2(p_n, q_n^Y) = 0. \quad (6)$$

Indeed, each interval $(c - \frac{1}{2}, c + \frac{1}{2})$ is mapped back to c . Thus a continuous law can match every discrete count probability at the chosen resolution. This construction copies the exact law and explains none of it. The useful goal is a compact model that predicts *unseen* sizes. Without a shared rounding rule, ordinary KL divergence between a discrete law and a continuous one is infinite, regardless of how closely their CDFs track each other.

A structural route to closing the useful gap

Let Z_m be independent, centered, unit-variance approximations to the standardized child count laws, and set $s_m = 0$ when $m \leq 2$. Restore the finite-size split $I \sim \text{Unif}\{0, \dots, n-1\}$:

$$\tilde{C}_n = n - 1 + \mu_I + s_I Z_I + \mu_{n-1-I} + s_{n-1-I} Z'_{n-1-I}. \quad (7)$$

Conditioning on I and using the exact mean and variance recurrences gives

$$\mathbb{E}\tilde{C}_n = \mu_n, \quad \text{Var}(\tilde{C}_n) = \text{Var}(\mu_I + \mu_{n-1-I}) + \mathbb{E}(s_I^2 + s_{n-1-I}^2) = s_n^2. \quad (8)$$

This moment guarantee does not force the tails to match. The table shows the gain when Z_m is based on Q , with exact small children at $n = 16$.

There is also a simple error guarantee for a proposed *count-level* version. Let P_m be the exact child law and R_m any discrete approximation (for example, a rounded smooth child law). Form $\tilde{P}_n$ from (1) with R_i, R_{n-1-i} in place of the true child laws. Convolution cannot increase total variation, so

$$d_{\text{TV}}(P_n, \tilde{P}_n) \leq \frac{2}{n} \sum_{m=0}^{n-1} d_{\text{TV}}(P_m, R_m). \quad (9)$$

This bound concerns the count-level variant; the table used continuous children followed by rounding the parent. Both suggest the next test: improve child laws by size, then evaluate the resulting parent at held-out sizes.

A different loss: forgetting the growing tree

The same comparison count can arise from different binary search trees. Under the natural growing-tree coupling, let $K_n(d)$ be the number of empty insertion slots at depth d , and let D_{n+1} be the depth chosen next. Then

$$C_n = \sum_d dK_n(d) - 2n, \quad \Pr(D_{n+1} = d \mid K_n) = \frac{K_n(d)}{n+1}. \quad (10)$$

Knowing only C_n is like knowing the total cost while forgetting where the remaining slots are. Two trees with the same cost can give different chances for the next insertion depth. The exact expected predictive penalty, in bits, for retaining only cost is

$$L_n = I(D_{n+1}; K_n \mid C_n) = H(D_{n+1} \mid C_n) - H(D_{n+1} \mid K_n). \quad (11)$$

Equivalently, L_n is the smallest expected extra log-loss of any predictor that sees only C_n , relative to one that sees K_n :

$$\inf_{a(\cdot \mid C_n)} \mathbb{E} D_{\text{KL},2}(\Pr(D_{n+1} \in \cdot \mid K_n) \parallel a(\cdot \mid C_n)) = L_n. \quad (12)$$

The minimizing predictor is $\Pr(D_{n+1} \in \cdot \mid C_n)$. This result concerns the natural tree coupling, not a particular smooth curve.

Exact enumeration gives $L_{16} = 0.31545$ bits. The largest value through $n = 26$ occurs at $n = 14$, about 0.31866 bits, followed by a decline to $L_{26} = 0.27815$. Two independent large-tree simulations estimate roughly 0.06 bits at $n = 512$. The evidence points toward a shrinking one-step gap, but does not prove that it vanishes. Meanwhile, the number of hidden profile details can grow: $H(K_n \mid C_n)$ rises from 2.38 bits at $n = 14$ to 5.85 bits at $n = 26$. More of the tree is hidden by the total cost, yet less of that hidden detail helps predict the very next insertion.

What “close the gap” means

The two bit values answer different questions. Equation (5) tests *one-time distributional fit* after declaring the count resolution. Equation (11) tests *next-step prediction* after compressing the tree to one number. They cannot be added or compared as parts of a single information budget. The exact continuous lift in (6) makes the distinction formal: because C_n can be recovered from Y_n and U_n supplies no tree information,

$$\text{JS}_2(p_n, q_n^Y) = 0 \quad \text{while} \quad I(D_{n+1}; K_n \mid Y_n) = I(D_{n+1}; K_n \mid C_n) = L_n. \quad (13)$$

At $n = 16$, the first quantity is zero and the second is 0.31545 bits.

For the first goal, take each Z_m from the standardized fractional family at generation $t_\beta(m)$, then let $q_{n,\beta}$ be the rounded parent law produced by (7). Fit the shared size-to-generation map on some sizes and test it on others:

$$\begin{aligned} \hat{\beta} &= \arg \min_{\beta} \frac{1}{|\mathcal{N}_{\text{train}}|} \sum_{n \in \mathcal{N}_{\text{train}}} \text{JS}_2(p_n, q_{n,\beta}), \\ \mathcal{E}_{\text{test}}(\hat{\beta}) &= \frac{1}{|\mathcal{N}_{\text{test}}|} \sum_{n \in \mathcal{N}_{\text{test}}} \text{JS}_2(p_n, q_{n,\hat{\beta}}). \end{aligned} \quad (14)$$

Skewness and right-tail probabilities should be checked alongside this score. For dynamics, the open theoretical question is the large- n behavior of L_n . The key modeling choice is how much *state* to retain, alongside how closely a smooth law fits the count distribution.

Reproducibility. The companion paper source, `quicksort_*.py` experiment scripts, and dated `artifacts/` notes in this repository contain the recurrence calculations, simulations, and limitations behind these figures.